

## Multiple Choice (10 marks)

1. Which living organisms least resemble the common ancestor of all life according to genetic testing?
  - (a) viruses
  - (b) bacteria such as E. coli
  - (c) organisms living deep in the oceans around seafloor volcanic vents and in hot springs
  - (d) plankton that use sunlight as an energy source through photosynthesis
2. Which of the following is/are NOT caused by gravitational interactions?
  - (a) 2:3 periodic ratio of Neptune:Pluto
  - (b) Kirkwood Gaps.
  - (c) Gaps in Saturn's rings.
  - (d) Breaking of small Jovian moons to form ring materials.
3. You cool a blackbody to twice its original temperature. How does its spectrum change?
  - (a) Power emitted is 1/16 times as high; peak emission wavelength is 1/2 as long.
  - (b) Power emitted is 1/4 times as high; peak emission wavelength is 2 times longer.
  - (c) Power emitted is 1/4 times as high; peak emission wavelength is 1/2 as long.
  - (d) Power emitted is 1/16 times as high; peak emission wavelength is 2 times longer.
4. A sand bag has a mass of 5kg and weight 50N on Earth. What is the mass and weight of this sand bag on a planet with half the gravity compared to Earth?
  - (a) Mass 5kg weight 100N
  - (b) Mass 5kg weight 50N
  - (c) Mass 5kg weight 25N
  - (d) Mass 10kg weight 100N
5. How do astronomers think Saturn generates its internal heat?
  - (a) nuclear fusion in the core
  - (b) by contracting changing gravitational potential energy into thermal energy
  - (c) internal friction due to its high rotation rate
  - (d) chemical processes

## True / False (10 marks)

*Answer True or False*

6. True or False: Betelgeuse is a red supergiant star situated in the constellation Cassiopeia, often mistaken for one of the brightest stars.
7. True or False: A scientific theory is considered valid after a single successful experiment that supports the hypothesis.
8. True or False: All of the jovian planets, including Neptune, Uranus, and Saturn, lack ring systems.
9. True or False: Mars exhibits evidence of surface processes including impacts, erosion, and volcanism.
10. True or False: The acceleration of the truck on Mars would be different from Earth, assuming no friction, due to the uniform application of force.

## Long Form Response (20 marks)

*Answer each question with a clear and complete explanation.*

11. Discuss the significance of the Laniakea supercluster within the context of the Milky Way's location and its implications for our understanding of cosmic structure.
12. Discuss the implications of Pluto's orbital period of approximately 250 years on its geological and atmospheric characteristics, considering how such a lengthy orbit influences seasonal changes and climate.

*End of Paper*